

The London Register of Subsurface CO₂ Storage

Methodology

Version 1.3 (data release v8)

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Document control

This document describes the methodology used to compile, process, quality-control, and publish the London Register of Subsurface CO₂ Storage (the “Register”). It is intended to be published alongside the Register release package (database and tabular exports) and the Annual Report.

Roles and responsibilities

The Register is maintained through an annual update cycle. The roles below describe the minimum responsibilities required to ensure consistent application of this methodology, traceability of values, and quality control prior to publication.

Role	Core responsibilities	Key outputs / records
Register Director / Methodology Owner	Owns this methodology; defines scope and conventions; approves any methodology or convention changes; resolves complex source conflicts; independently cross-checks extracted values and transformations against the archived source (Four-eyes for QC); verifies application of rules; confirms citations and register placement; signs off publication	Approved methodology version; prioritisation decisions; publication sign-off; issue tracking records.
Register Executive Director / Technical Maintainer	Identifies relevant sources; archives source files; extracts raw values and supporting context; records all extraction details and assumptions; Maintains the data repository structure (folders, naming, access); supports automation and quality checks where implemented; maintains machine-readable exports.	QA record entries; archived source files; raw value records; reviewer sign-off; correction actions raised and closed.
Project Manager	Prepares the published Register package; validates file naming and version identifiers; maintains the release change record; ensures publication and archiving of releases.	Published Register package; release change record; archived releases.

Version history

This document is versioned. Substantive changes to methodology, processing rules, or conventions are recorded below.

Version	Date	Summary of changes
1.0	2026-07-10	Publication aligned with the DNV data quality assurance recommendations; includes workflow, processing rules, source prioritisation, QC approach, and reporting standards context.
1.1	2026-09-17	Data release v8: metric-basis fields (metric_basis, metric_basis_notes) added to raw_observation and documented in Sections 4.6, 4.7.4, Appendix A and Appendix C; data-dictionary field names aligned to the database; Figure 4-1 updated to show all raw_observation fields; cross-references corrected (Section 2.2 now points to Section 11; Section 3 to Section 7.2); note added that publication_date is stored as TEXT in the SQLite export.
1.2	2026-09-18	V8 source audit: historical Norway observations rebound to NID sources with the official 2021 correction preserved; direct Croatia 2020 annual observation and cumulative residual handling documented; Weyburn source and 2018 transcription corrected; complete metric-basis evidence retained; fiscal-year labels, cumulative chaining and portable SQLite reproduction clarified.
1.3	2026-09-22	Final v8 review: FOIA quantities aligned to Subpart RR net sequestration; database-first notebook workflow, published artefacts, selection tiers, review coverage, definitions and field formats clarified. Existing quantities and processing conventions retained.

1. Introduction

The London Register of Subsurface CO₂ Storage compiles project-level annual amounts of CO₂ stored underground by operational CO₂ storage projects worldwide. The Register aims to provide a consolidated, traceable dataset designed to support scientific research, policy-making, investment decisions, and public transparency.

This methodology document provides a description of: (i) what is included in the Register; (ii) how projects and data sources are identified; (iii) how values are extracted and transformed into a consistent format; (iv) how multiple sources are handled; (v) how quality control is performed; and (vi) how versions are managed and published.

1.1 Audience

This document is written for Register users (researchers, analysts, policy-makers, and other stakeholders) as well as contributors and maintainers who participate in the annual update process.

1.2 Relationship to other project materials

This document should be read together with: (a) the published Register database (which contains the data as a relational database); and (b) the Annual Report (which provides headline trends and visualisations derived from the Register). An independent data quality assurance review has been conducted by DNV; recommendations from that review have informed the structure and content of this methodology.

1.3 Data quality principles

The Register is designed and maintained in accordance with four overarching data quality principles as outlined in a third-party assessment, specifically referenced as 'the independent third-party data quality assurance evaluation of the Register.'

Completeness: The dataset aims to capture all relevant operational subsurface CO₂ storage projects and their reported annual storage values, within the defined scope.

Understandability: Users should be able to interpret values correctly, supported by clear definitions, notes, and documented conventions.

Traceability: Each value should be traceable back to an underlying source, with an audit trail of any transformations applied.

Accuracy (representativeness): Accuracy is framed against the representativeness of the available sources and the transparency of assumptions and transformations.

2. Scope and boundaries

2.1 Temporal and geographic scope

The Register covers annual CO₂ storage amounts from 1996 to the latest reporting year available at the time of publication. The Register is global in scope and is updated on an annual cycle.

Some entries for the most recent year may be incomplete due to reporting delays; where so, missingness is explicitly represented rather than imputed.

2.2 Included projects and reported variable

The Register focuses on CO₂ quantities associated with operational geological storage projects. Sources may report stored, injected, or received quantities; their accounting basis and any compiler derivation are recorded in the observation notes and metric-basis fields (Sections 4.7.4 and 6.3). Definitions and reporting frameworks are discussed in Section 11.

Where an operator publishes a company-level "captured for storage" volume that cannot be attributed to individual sites with verified records, the Register may include that data as an aggregate accounting entity (e.g., ExxonMobil Aggregate Projects). An aggregate entity is not a single site, so its storage type is left blank, its scope is documented in the project notes, and its priority values are netted off all Register projects in which the reporting company participates (see Section 7.8) so that no volume is counted twice.

2.3 Exclusions

The Register excludes planned projects. Capture, transport and utilisation metrics are included only where the documented record relates them to an operational storage project or aggregate accounting entity within the scope above.

3. Definitions and terminology

Key terms used in this document are defined below:

Aggregate accounting entity: A Register record representing a company-level disclosure spanning several operations, used when volumes cannot be attributed to individual sites; carried with explicit netting rules to avoid double counting.

Back revision: An update to previously published years when new data becomes available, or corrections are required.

Cumulative reporting basis: A source practice of reporting a running total stored since start-up rather than annual amounts; handled by differencing (Section 7.2) when the basis is confirmed.

Data source: The document, database record, or direct operator communication from which a value or metadata field is obtained.

MtCO₂/yr: Million metric tonnes of CO₂ per year. The standard unit used in the Register.

Primary source: A source reported directly by the operator (e.g., ESG/sustainability report or survey response) or an official government/regulator dataset.

Processed value: A project-year value derived from selected raw observations by the unit conversion, annual allocation and deduction rules in Section 7.

Project: An operational CO₂ storage project or storage operation at a site for which annual subsurface CO₂ injection or storage is reported.

Project-year value: An annual amount selected or derived for a project under the processing rules in Section 7, with the source metric and any compiler treatment retained in the notes.

Raw observation: The database record of a source value or a documented compiler-derived quantity, with its unit, reporting period and evidence. Original source statements and the steps behind any derived quantity are retained in the notes.

Register: The London Register of Subsurface CO₂ Storage dataset, maintained in a relational database and published annually with associated user products. The Annual Report and independent assurance reports are accompanying publications.

Secondary source: A third-party compilation such as an industry report, research publication, or environmental assessment used when primary data are unavailable.

The database: The SQLite release file containing the four linked core tables and the views used to derive project-year values.

User products: The stakeholder spreadsheet and figures generated from the database for reporting and analysis.

4. Dataset structure and how to read the Register

4.1 Published file structure

The v8 release contains the SQLite database, CSV exports of its four core tables, SQL Server build and query scripts, one self-contained Jupyter notebook, the stakeholder spreadsheet, four PNG figures, this methodology and the reference archive. The database is the authoritative calculation input; the CSV files provide matching tabular copies with consistent identifiers.

4.2 One row per project-source combination

The Register retains multiple sources when available. From 2026 onwards, the master dataset records one row per raw extracted observation (raw_observation), linking a project, a source, and a release. Multiple sources for the same project and year are represented by multiple raw_observation records, preserving transparency and enabling comparison.

4.3 Core fields

The information is stored across four linked tables: log (release metadata), project (project metadata), source (source metadata and archiving reference), and raw_observation (reported values, units, periods and documented compiler quantities). The v_headline_series view derives project-year values in MtCO₂/yr; the self-contained notebook builds the user products from that view (Sections 6 and 7).

4.4 Unique identifiers

Each release is identified by `log_id` (e.g., 2026_v1). Projects and sources use stable internal identifiers (`project_id`, `source_id`). Each extracted record is identified by (`raw_id`, `log_id`), where `raw_id` is generated deterministically from project identifier, reporting period, and priority tier to support duplicate detection and quality control.

4.5 Presentation conventions

Published exports (CSV, and any optional spreadsheet views generated for convenience) avoid relying on formatting for meaning. Conventions are communicated through explicit fields and documented codes (e.g., `period_type`, `priority_tier`) plus notes where needed.

4.6 Data dictionary (summary)

Field / column	Description	Type	Notes / examples
project_name	Project name used in the Register (earlier label: CO ₂ Storage Project)	Text	E.g., Sleipner
storage_operator	Operator responsible for storage operations (earlier label: Storage Operator)	Text	
storage_type	Storage context	Categorical	E.g., Geological Storage; Geological Storage with Oil Recovery
Starting year	First year of injection/storage operation	Numeric	Not a database field in this release; recorded in <code>project_notes</code> where known
co2_source	Origin of CO ₂ stream, where available (earlier label: CO ₂ Source)	Text	E.g., natural gas processing
source_name / url / archived_name	Reference and archive handle for the reported values (earlier label: Data Source); held in the source table	Text	Public URL where available; archived filename always recorded
Year columns (user products)	Annual CO ₂ storage values under the documented reporting-year labels, derived from <code>raw_value</code> , <code>raw_unit</code> and the reporting period under the Section 7 rules	Numeric	Standard unit: MtCO ₂ /yr

Field / column	Description	Type	Notes / examples
standards_text	Reporting frameworks explicitly cited by the source (earlier label: Reporting standard flags)	Text	E.g., EPA GHGRP Subpart RR; GRI
raw_observation_notes / project_notes / source_notes	Notes on assumptions, conversions and context (earlier label: Note)	Text	Includes annualisation details
metric_basis; metric_basis_notes	Metric basis of the stored quantity as defined by the source, with source evidence and locators	Categorical; Text	Added in data release v8; covers the audited EOR, geological-storage and aggregate observations described in Section 4.7.4

Field names below are the relational database fields (release v8). Labels used in earlier spreadsheet editions are noted in the description where they differ.

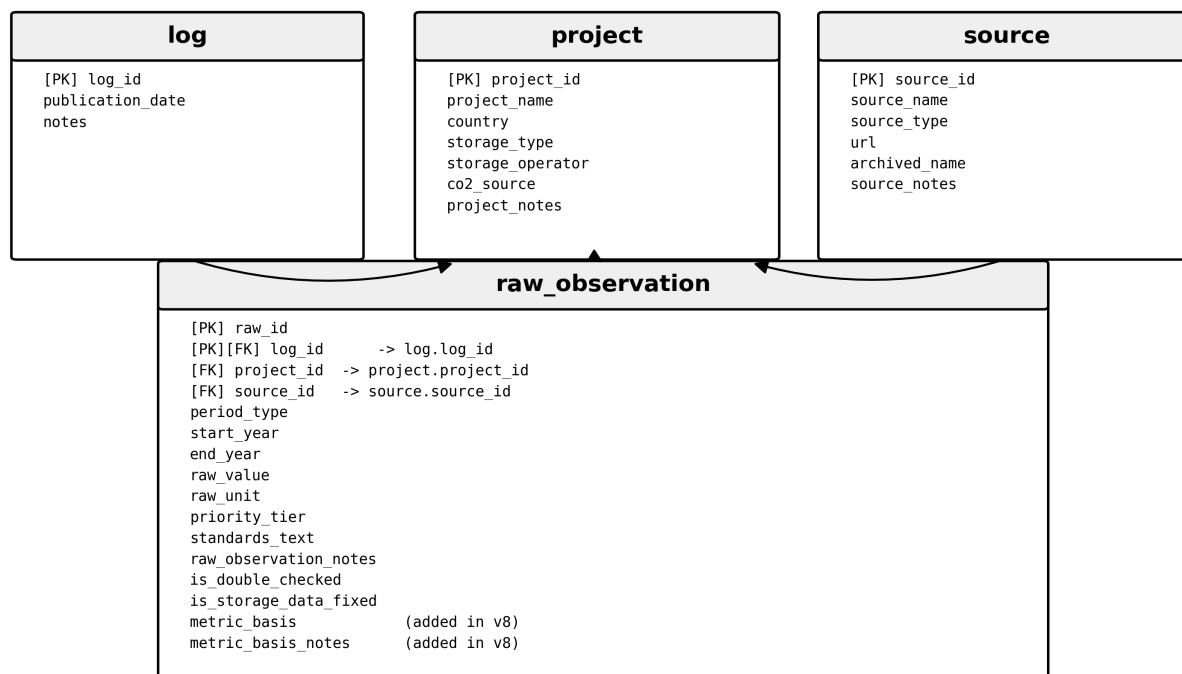
4.7 Relational database structure (2026 onwards)

From 2026 onwards, the Register is maintained as a relational database to improve traceability, reduce duplication, and enable consistent machine-readable exports. The database stores (i) release metadata; (ii) stable project metadata; (iii) source metadata and archiving references; and (iv) source observations and explicitly documented compiler-derived quantities. The detailed schema is provided in Appendix C.

4.7.1 Core tables and relationships. The database uses four core tables: log (release metadata), project (project metadata), source (evidence records), and raw_observation (raw extracted values). Each raw_observation row references exactly one log, one project, and one source via foreign keys. This structure preserves multiple sources per project-year and supports auditability.

Table	Purpose	Primary key
log	One row per Register release/version (publication date, notes).	log_id
project	One row per operational CO ₂ storage project (stable metadata).	project_id
source	One row per source document/database record (URL + archived reference).	source_id
raw_observation	One row per extracted observation linking project, source, and release.	(raw_id, log_id)

4.7.2 Schema diagram. Figure 4-1 summarises the core tables and key relationships used to store the Register data.



raw_observation is the fact table (one extracted record). log = release/version; project = storage project; source = evidence record.

Figure 4-1. Core relational database schema used for the 2026 Register (data release v8) (PK = primary key; FK = foreign key).

4.7.3 Identifier conventions. Identifiers are designed to be stable within a release. **log_id** follows the format 'YYYY_vN' (e.g., 2026_v1). **project_id** and **source_id** are stable internal identifiers. **raw_observation** uses a composite primary key (**raw_id**, **log_id**). **raw_id** is generated deterministically from a project slug, the reporting period, and the priority-tier suffix to help detect duplicates; the current implementation follows the pattern: raw-<project_slug>-<start_year>-<end_year>-<priority_tier>.

4.7.4 Controlled values and required fields. The **raw_observation** table records source values or documented compiler quantities in **raw_value** and **raw_unit**. **period_type** is coded as 1 (single-year) or 2 (multi-year). **priority_tier** is coded as 1 (selected for headline processing) or 2 (retained alternative). These selection codes are distinct from primary and secondary source provenance: either tier can contain operator or government records. **standards_text** records frameworks explicitly cited by the source.

raw_unit is restricted to Mt, kt and tonnes. **period_type**=1 requires **start_year** = **end_year**; **period_type**=2 requires **start_year** < **end_year**. The **raw_id** suffix equals **priority_tier**. Review flags are **is_double_checked** and **is_storage_data_fixed** (0/1). **metric_basis** uses program_net, net_stored, gross_injected, co2_received or not_stated; **metric_basis_notes** retains source quotations, locators and compiler interpretation in full. In v8, 246 observations have this classification: 202 EOR-associated, 26 geological-storage and 18 aggregate observations. Blank fields indicate records outside the completed classification coverage. **program_net** identifies Subpart RR net sequestration reporting with an approved MRV plan. The 2024 EIP/FOIA records

use the same RR annual sequestered-mass basis as earlier EPA data; the map summary heading “Injected CO₂” does not redefine that quantity.

5. Project identification and source discovery

5.1 Building and maintaining the project list

Projects are identified using: (i) global reports and databases of operational CO₂ storage projects; (ii) input from the Register consortium and expert reviewers; and (iii) known operators and jurisdictions where CO₂ storage is reported. The project list is reviewed during each annual update cycle to add newly operational projects and to identify projects that have ceased operation or changed reporting practices.

5.2 Source discovery and types

Annual CO₂ storage amounts are gathered from multiple source types. Primary sources include: (i) data reported directly by project operators (e.g., sustainability/ESG reports, annual reports, or survey responses), and (ii) official government or regulator datasets (e.g., national inventories or regulatory reporting databases). Secondary sources, such as industry reports, research publications, or environmental assessments, may be compiled particularly when primary data are unavailable.

5.3 Source archiving

To support long-term traceability, sources are archived by downloading and saving copies to the project’s internal file storage. Archived sources are stored with structured naming (including project name, year, source type, and retrieval date) so that each Register value can be traced back to an immutable copy even if the original URL changes or is removed. This archive may be made publicly available as a part of the Register.

6. Data gathering and update process

The Register is updated on an annual cycle using a repeatable workflow. The workflow combines manual review of source documents with structured documentation and quality control steps.

6.1 Overview of the workflow

At a high level, the update workflow is defined by the following steps.

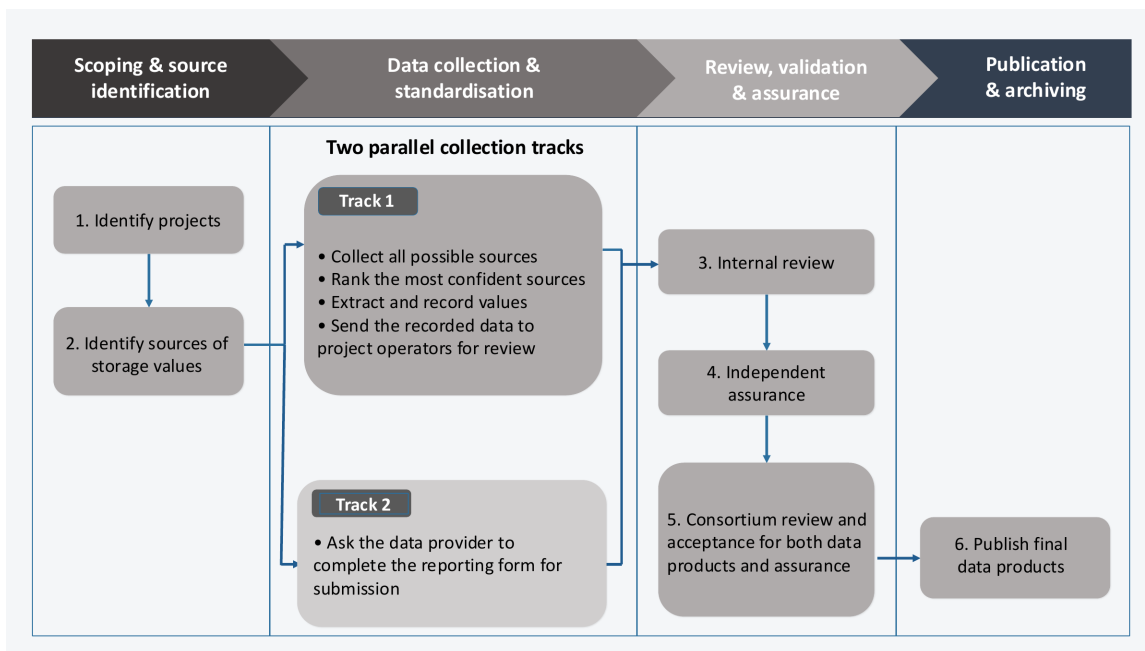


Figure 6-1 End-to-end workflow for data sourcing, validation, assurance, and publication

6.2 Detailed update workflow

1. Review and update the project list for the new cycle (add newly operational projects; retire or merge records where justified) and ensure each project has a stable project_id.
2. Identify candidate data sources for each project and reporting year(s); create or update source records (source_id, source_type, url).
3. Archive each source (download and store an immutable copy) and record the retrieval date and archived_name/handle in the source record.
4. Extract source values with their units, reporting periods and page/table references. Where a compiler calculation or release convention is needed, record the source statement, inputs and treatment in raw_observation_notes or metric_basis_notes.
5. Insert or update observations for the target log_id, assigning priority_tier and applicable standards_text. Routine mass-unit conversion and annualisation occur in the processed view; any calculation already embodied in a raw record is explicitly documented.
6. Perform quality control: automated checks (schema integrity, duplicates, unit and period validity, annualisation re-summing) and four-eyes spot checks against archived sources.
7. Resolve any issues by correcting the raw database records (or source/project metadata) and re-running the processing pipeline until checks pass.
8. Package and publish the database, four table CSVs, SQL scripts, notebook, stakeholder spreadsheet, four figures, methodology and reference archive, then archive the release.

6.3 Recording raw versus processed values

The workflow distinguishes source evidence, database observations and processed project-year outputs. Some observations already contain documented calculations: Croatia 2014–2015 converts reported gas volume using the source density; North Burbank Unit 2013–2019 deducts produced/recycled CO₂ from injected CO₂. The retained Otsego annual allocation and Petra Nova 2021–2022 zero-value convention are also identified in notes, with the underlying source evidence preserved. The v_headline_series view applies the subsequent unit conversion, cumulative allocation and deductions. The notebook reads that view and reproduces the user products independently of earlier releases.

6.4 Documentation required for each update

Each update maintains the four core tables, source archive, observation notes and release log. The v8 publication package is specified in Section 10.5; its notebook exports the four CSV tables and regenerates the stakeholder spreadsheet and four PNG figures directly from the SQLite database.

6.5 Conventions used in published exports

Published exports (CSV, and any optional spreadsheet views generated for convenience) use explicit fields rather than formatting to communicate key information. Examples include: period_type (single-year vs multi-year cumulative), clear notes for any annualisation assumptions, explicit indicators of prioritised sources for headline statistics, and standardised representations for missing values. All conventions and codes used in a given release must be defined in the release documentation and remain consistent across files.

7. Data processing and harmonisation rules for user products

This section describes how selected database observations are standardised and interpreted for user products. Original source statements remain traceable through the archived files and evidence notes, including where a database observation contains a compiler-derived quantity (Section 6.3).

7.1 Unit normalisation

All annual values in the Register are standardised to MtCO₂/yr. If a source reports values in other mass units, values are converted using standard metric conversions and recorded with appropriate significant figures.

Common conversions (metric):

Reported unit	Convert to MtCO ₂ /yr	Example
t CO₂ (metric tonnes)	value / 1,000,000	250,000 t = 0.25 Mt
kg CO₂	value / 1,000,000,000	1.2e9 kg = 1.2 Mt
kt CO₂	value / 1,000	450 kt = 0.45 Mt

If a source uses non-metric tons or a volume unit, the conversion factor and original quantity must be documented in the observation notes before recording a value in the controlled mass units.

7.2 Annualisation of multi-year cumulative values

Some sources report cumulative CO₂ stored over a multi-year period (e.g., “X Mt stored from 2018-2020”). In these cases, the Register derives an annualised average by dividing the cumulative amount equally across the stated years, unless the source provides year-by-year breakdowns or another allocation method.

Formula (default):

$$\text{Annualised value for each year in period} = \frac{(\text{Cumulative value})}{(\text{Number of years})}$$

The reporting period and the annualisation assumption are documented in the notes and in the derived processed outputs.

Additional rules:

- (a) Overlap netting: where annual priority records already exist for years inside a cumulative period, those annual values are subtracted from the cumulative total and only the residual is distributed evenly across the uncovered years.
- (b) Cumulative chaining: where cumulative records share the same project and start year, the processing pipeline orders the checkpoints by end year and differences consecutive totals. Each increment is allocated only within its new year segment after subtracting any directly reported annual observations in that segment.
- (c) Basis and period checks: cumulative checkpoints and annual observations are checked for compatible quantities and boundaries. Derived allocations are recorded in the notes. The v8 decision retains Core Energy Otsego’s existing five annual allocations; the CRS source’s December 2019 qualifier and the allocation basis remain documented in the evidence notes.

7.3 Calendar year, fiscal year, and reporting period alignment

The Register uses annual year labels. Calendar-year sources retain their stated year. For the Gorgon series, a financial year spanning two years is assigned to its starting-year label (for example, 2024/25 is recorded under 2024); the period is retained in the observation notes. These labels do not convert fiscal-year quantities into calendar-year measurements. Partial-year and cumulative-to-date disclosures remain identified as such, without extrapolation to a full-year amount.

7.4 Handling multiple sources and discrepancies

Where multiple sources provide values for the same project-year, each source is retained in a separate project-source row. Discrepancies are documented in notes. The Register avoids undocumented reconciliation. Where a single value is required for aggregation, source prioritisation rules (Section 8) are applied and documented.

7.5 Precision and rounding

Source precision and documented compiler rounding are retained in the database. Calculations use unrounded numeric values. The stakeholder spreadsheet displays annual project and

country amounts to three decimal places in Mt, its hidden chart-data sheet to four, and headline summaries generally to one. Display rounding does not alter the underlying calculations.

7.6 Missing values and null indicators

Missing annual quantities are left blank. The processing rules allocate reported cumulative totals only over their documented periods; they do not extrapolate unreported years. The four CSV exports use empty fields for nulls, and the stakeholder spreadsheet leaves missing values blank.

7.7 Double counting and internal consistency checks

To reduce the risk of double counting or misalignment (e.g., a single-year value inadvertently attributed to multiple years), the update process includes cross-checks, including verifying that annualised values sum back to the reported cumulative value; ensuring that values are not duplicated across adjacent years without a documented basis; and confirming consistency with project start year.

7.8 Company-level aggregate values and double-counting avoidance

Where an aggregate accounting entity carries a company-level captured-for-storage disclosure, its priority value for a year is the company figure minus the full priority-1 values of all Register projects in which the reporting company participates in that year (for ExxonMobil: Shute Creek, Gorgon, and Qatar LNG).

8. Source prioritisation for user products

The Register database is designed to retain multiple sources for transparency. However, some user products (e.g., figures in the Annual Report) require a single selected value per project-year. In those cases, the following prioritisation approach is applied.

8.1 Prioritisation principles

- Prefer primary sources over secondary sources.
- Prefer sources with clear definitions of the reported metric (injected/stored, boundaries, and reporting period).
- Prefer official regulatory/government datasets where they provide site-level annual values under defined methodologies.
- Prefer the most direct and most recent disclosure for the relevant year (e.g., a site report over a high-level corporate summary).
- Where an operator provides values via a survey, treat this as a primary source when accompanied by sufficient context and/or supporting documentation.
- Prefer site-level regulatory reporting (e.g., EPA GHGRP Subpart RR) over company-level corporate metrics for site records; carry company-level metrics only through aggregate accounting entities (Section 7.8).
- Prefer full-precision operator-report values over rounded regulator summaries where both describe the same quantity.
- Treat responses to formal government information requests (FOI/RFI, e.g., the Norwegian Environment Agency) as primary sources; the archived response (document, email, or web page) is the archived source.

- Where a multi-source survey compilation (e.g., a GCCSI survey) is used, record the underlying provenance of each value (regulator website, company filing, direct provider report) in the observation notes.

8.2 Documenting prioritisation decisions

When a prioritised source is selected, the rationale is recorded in the dataset notes. If sources conflict materially and there is no clear basis to resolve the discrepancy, multiple values are retained.

9. Quality assurance and quality control

Quality assurance for the Register is based on transparency, traceability, and repeatability. The project combines structured review, automated checks where feasible, and consortium oversight.

9.1 Four-eyes review

The four-eyes process asks a second team member to check transcription, unit conversion and annualisation, source references and notes, and application of the release conventions. Review completion is recorded per observation rather than inferred from inclusion in the release.

The v8 database records `is_double_checked=1` for 397 of 490 observations. Of the 399 priority-1 observations, 63 have a zero flag. A zero indicates that completion is not recorded in this field; it does not establish whether other review took place. `is_storage_data_fixed` separately records confirmation after review or correction.

9.2 Processing and QA record

A lightweight processing and QA record is maintained to support the four-eyes review and reproducibility. It focuses on accountability and non-standard decisions (e.g., ambiguous units, reporting-period interpretation, or back revisions). Routine transformations are reproducible from the database and the versioned processing pipeline, and do not require a separate public template.

9.3 Data quality checks

Prior to publication, the team performs checks aligned with common data quality categories (syntactic, semantic, and pragmatic). Indicative checks include:

- Uniqueness of project-source identifiers (no duplicate rows with the same composite key).
- All numeric values are valid numbers and non-negative.
- Values are consistent with the stated starting year (no values before the start year unless documented).
- Where annualisation is applied, the derived annual values re-sum to the reported cumulative value (within rounding).
- Each row has a populated Data Source reference (and, where possible, page/table location).
- Final checks include valid foreign-key links across the four tables, uniqueness of (`raw_id`, `log_id`), valid periods (annual records have `start_year` = `end_year`), and no duplicate priority-1

records for the same project, period and period type. Source records may be retained for corroboration without a selected observation.

Reproducibility is verified by executing the public notebook with the released SQLite database as its only project-data input and comparing the regenerated four CSVs, stakeholder spreadsheet and four figures with the published products.

10. Traceability, versioning, and publication

10.1 Master file and published releases

From 2026 onwards, the compiled data are maintained in a relational database. The v8 release provides a SQLite implementation with the four authoritative tables and derived views. Each release has a unique log_id and an archived evidence package so historical observations, corrections and source versions remain accessible.

10.2 File naming and identifiers

The v8 directory and release filenames identify the dataset version. The database and four CSV filenames also retain their existing 0917 suffix; the log table records the release publication date. Methodology revisions carry a separate version and date in this document.

10.3 Change management and back revisions

The Register may be revised retrospectively when new data become available, operators update earlier figures, or errors are identified. The log table and observation notes record what changed, why, and which projects or years are affected. Appendix B describes the release change record.

Material changes in headline figures are traced to interpretation changes, source updates or coverage additions through those records. The v8 package carries the release history in log_v8_0917.csv and the database; no separate reconciliation workbook is required to reproduce its products.

Currently back revisions are only published annually with subsequent publication cycles of the Register, i.e., the Register is not updated in between publication cycles.

10.4 Source volatility and long-term access

Because some public URLs change or become inaccessible over time, the Register maintains an internal archive of source documents. Where feasible, the Register also records persistent identifiers (e.g., DOIs) or stable landing pages, and notes when a source link has been updated.

10.5 Database release artefacts and exports (2026 onwards)

The published v8 package comprises Register/co2register_v8_0917.db, the four table CSVs, the SQL Server build and query scripts under Register/sqlcode_0917, BUILD_V8_FROM_DB.ipynb, the stakeholder spreadsheet and four PNG figures under products,

this methodology, and reference-v8. The SQL build script creates a separate SQL Server implementation; it is not a SQLite rebuild script.

Release identity. The database and CSVs retain log_id 2026_v8 for the current observations. Existing filenames are kept stable, and the log table preserves the dated release history. Methodology version 1.3 documents the final review corrections without changing the data-release identifier.

Export definition. The notebook copies all fields and rows of log, project, source and raw_observation to their four CSV files. Their identifiers preserve the joins between evidence, project metadata and release history. A user can query a joined view, but no denormalised CSV is part of this package.

Reproduction. Run the notebook from v8 with the supplied SQLite database and the installed dependencies described at its start. It exports the four CSVs and regenerates the stakeholder spreadsheet and all four PNG figures. Its code and chart assets are embedded; no external project JSON or Python file is needed. It does not rebuild the database or generate this methodology document. The SQLite database remains the authoritative input, and the CSVs are its matching exports.

11. Reporting standards and quantification context

Some sources referenced by the Register cite sustainability reporting frameworks or regulatory programs that influence how CO₂ storage is defined, measured, and disclosed. The Register records these frameworks as metadata where they are explicitly referenced.

11.1 Why reporting standards are included

Reporting standards are included to provide context for users. They may help interpret differences between projects in: (i) the boundary of the reported metric (e.g., injected vs stored); (ii) definitions of storage; (iii) requirements for monitoring, reporting, and verification; and (iv) assurance or audit expectations.

However, reporting standards vary in scope, level of prescription, and whether they are voluntary or mandatory. There is not necessarily a straightforward relationship between a cited reporting standard and the confidence that can be placed in a specific project-year value; users should consult the underlying source and, where applicable, the governing regulation.

11.2 Standards commonly referenced in the Register

IPIECA CCE-3 indicator: Oil and gas sector sustainability reporting guidance (voluntary).

GRI (Global Reporting Initiative): Cross-sector sustainability reporting framework with principles for completeness and accuracy (voluntary).

SASB Standards: Sector-specific disclosure standards aimed at financially material ESG information (voluntary).

HKEX ESG Reporting Guide (KPIs): Mandatory ESG disclosure requirements for Hong Kong listed companies (mandatory for listed entities).

U.S. EPA GHGRP (including Subpart RR where applicable): Regulatory program specifying methodologies for reporting geologic sequestration in the United States (mandatory for covered facilities).

U.S. EPA UIC program (class/well information): Regulatory framework covering underground injection control; used for contextual identifiers where relevant.

NGER — Australian National Greenhouse and Energy Reporting Scheme: Mandatory Australian scheme for reporting greenhouse gas emissions, energy production and energy consumption. Used as regulatory context for Australian projects; project-level stored CO₂ values are used only where clearly reported.

Puro Standard — Geologically Stored Carbon methodology and General Rules: Independent certification framework for geologically stored carbon removals. Used only where verified carbon-removal storage records are relevant.

UNFCCC National Inventory reporting under the 2006 IPCC Guidelines: National inventory reporting framework. Used where national inventory documents provide evidence for CO₂ injection, storage or country-level accounting.

ISO 50001 Energy Management Systems: Energy management standard. Used as supporting context only; not a CO₂ storage accounting standard.

TCFD — Task Force on Climate-related Financial Disclosures: Climate-related financial disclosure framework. Used for company climate strategy and risk-disclosure context; not a primary storage-volume source.

API Compendium and IOGP Sustainability Reporting Guidance: Oil and gas sector GHG estimation and sustainability reporting guidance. Used to interpret company-reported emissions and CCUS disclosures.

11.3 How standards information is populated

A reporting standard flag is populated when the data source explicitly states that the project/company reports under the framework or uses the framework's indicators. If the framework is not mentioned, the field is left blank.

11.4 Interpreting reporting standard flags

Users should treat reporting standard flags as contextual information rather than as a direct measure of data accuracy. For example, EPA programs may provide more prescriptive, site-level reporting requirements, while corporate ESG frameworks may provide broader disclosure guidance that is not specific to subsurface CO₂ storage. The Register does not assign confidence scores based solely on reporting standards.

12. Contact and feedback

The project welcomes feedback and corrections. Please contact the Register team via the project website or the contacts listed in the Annual Report.

13. References

1. DNV (2025). Data quality assurance - London Register of Subsurface CO₂ Storage. Report No. 10591504-02, Rev. 1.
2. Gao, X., & Krevor, S. (2025). The London Register of Subsurface CO₂ Storage: 2025 Annual Report. Imperial College London.
3. DNV-RP-0497 (2023). Assurance of data quality management.
4. ISO 8000 series. Data quality.
5. ISO/IEC 25012:2008. Data quality model.
6. Gao, X., & Krevor, S. (2026). The London Register of Subsurface CO₂ Storage: 2026 Annual Report. Imperial College London.
7. Puro.earth. Puro Standard - Geologically Stored Carbon methodology and General Rules.
8. IPCC (2006). 2006 IPCC Guidelines for National Greenhouse Gas Inventories.

Appendix A. Data dictionary (detailed)

This dictionary summarises the released field meanings, controlled values, units and null handling. Appendix C gives the SQLite field types; the accompanying T-SQL script specifies the SQL Server types.

Field	Definition	Type / format	Allowed values / constraints	Examples / notes
storage_type	Categorisation of the storage context	Categorical	Geological Storage; Geological Storage with Oil Recovery; null	Aggregate accounting entities have a blank storage_type .
raw_value / raw_unit (with period_type, start_year, end_year)	Source amount or explicitly documented compiler-derived amount, with its unit and reporting period; project-year values follow Section 7	Text (numeric string) / controlled unit	raw_unit in {Mt, kt, tonnes}; value >= 0	Original evidence and any conversion, calculation or release convention are retained in the notes.
source_id → source (source_name, url, archived_name)	Traceable reference to the underlying source	Text (URL/citation)	Must be present for each row	Prefer a direct link to the specific document and year.
raw_observation_notes	Contextual notes for interpretation	Text	Free text	Use to document assumptions, allocation periods, discrepancies, and anomalies.
standards_text	Frameworks cited by the source	Categorical	Populated only when explicitly referenced	Used as context, not a confidence score.
is_double_checked	Four-eyes review completed for this observation	Integer (0/1)	0 or 1; added in data release v5	1 = reviewed by a second team member.
is_storage_data_fixed	Value confirmed final after review/correction	Integer (0/1)	0 or 1; added in data release v5	1 = value checked and fixed against the source.

Field	Definition	Type / format	Allowed values / constraints	Examples / notes
metric_basis	Metric basis of the stored quantity as defined by its source	Text (controlled)	program_net; net_stored; gross_injected; co2_received; not_stated. Added in data release v8	program_net = Subpart RR net sequestration reporting with an approved MRV plan. Coverage includes audited EOR, geological-storage and aggregate observations.
metric_basis_notes	Source quotations and locators, with compiler interpretation distinguished, supporting the metric-basis classification	Text	Free text with page, sheet or cell references. Added in data release v8	Quotes the source sentence or table column that carries the value.

Appendix B. Release change records

The v8 release records changes in the log table and supporting observation notes. The entries identify:

- Release identifier and date
- Summary of new projects added / removed
- Summary of new source documents added
- Summary of back revisions (project, years affected, reason)
- Summary of methodology/convention changes (if any)
- Known limitations for the release (e.g., expected reporting delays for the most recent year)

B.1 Purpose and principles

The release change record preserves the reason and scope of each update so users can trace historical revisions, understand reporting limitations and reproduce a specific dataset version. In v8 this record is contained in the database and log CSV.

B.2 Recommended release identifier and versioning

Use the release identifier consistently across the database records and published package. The v8 data-release identifier is 2026_v8; the methodology has its own revision history. The publication date in log is retained when explanatory material is corrected within the same release.

B.3 Release identity

The current release is identified by the following metadata. Detailed source and observation changes are preserved in the log table and evidence notes.

Field	Entry
Release identifier	2026_v8
Release publication date	18/09/2026
Database filename	co2register_v8_0917.db
Coverage years	1996–2025, with reported-to-date coverage in 2025
Maintainers	Dr Xiaowei Gao; Prof Sam Krevor
Contact	x.gao24@imperial.ac.uk; s.krevor@imperial.ac.uk

B.4 Methodology and convention changes

The methodology version history records changes to processing descriptions and release conventions. Observation notes distinguish source updates from compiler decisions; the log retains the affected projects, years and reasons. Final v8 review corrections align FOIA metric descriptions and the documented reproduction workflow with the released files.

Appendix C. Relational database implementation details (2026 onwards)

This appendix documents the database implementation used to store and export the Register from 2026 onwards. It complements Sections 4.7 and 10.5 by providing a concise, reproducible description of how data are ingested, stored, and exported in machine-readable form.

C.1 Database technology.

The v8 release includes a SQLite database with derived views, four matching CSV exports, and separate SQL Server T-SQL build and query scripts. The SQLite v_headline_series view applies unit conversion, cumulative chaining, overlap netting and company-aggregate deductions. One self-contained notebook reads this database and exports the four CSVs, stakeholder spreadsheet and four PNG figures. It does not build the database or author documents.

C.2 Data ingestion workflow.

Each annual update begins by creating (or updating) records in the four core tables: log, project, source, and raw_observation. Data can be entered directly into the database, or captured via a structured entry template that mirrors the database tables and then batch-loaded. During loading, identifiers are validated, required fields are checked for completeness, and raw_id is generated deterministically to reduce duplicate insertions.

C.3 Table schema summary (authoritative fields)

The table below lists SQLite types. The SQL Server script uses NVARCHAR for text, INT for integer fields and DATE for publication_date, with its declared lengths and constraints.

Table	Field	Type (SQLite)	Notes / constraints
log	log_id	TEXT	PK; not null; Release identifier (e.g., 2026_v1).
log	publication_date	TEXT	not null; DD/MM/YYYY in SQLite and CSV. The SQL Server script uses DATE with explicit conversion from YYYYMMDD.
log	notes	TEXT	nullable; Release history and comments.
project	project_id	TEXT	PK; not null; Stable project identifier.
project	project_name	TEXT	not null; Project name (as published).
project	country	TEXT	not null; Country.
project	storage_type	TEXT	nullable; Storage type / context (blank for aggregate accounting entities).

Table	Field	Type (SQLite)	Notes / constraints
project	storage_operator	TEXT	not null; Operator.
project	co2_source	TEXT	nullable; CO2 source (if known).
project	project_notes	TEXT	nullable; Project notes.
source	source_id	TEXT	PK; not null; Stable source identifier.
source	source_name	TEXT	not null; Short label / title.
source	source_type	TEXT	nullable; Category (e.g., Operator; Regulator).
source	url	TEXT	nullable; Public URL (if any).
source	archived_name	TEXT	nullable; Archived filename / handle.
source	source_notes	TEXT	nullable; Source notes / citations.
raw_observation	raw_id	TEXT	PK (composite); Derived extracted-record ID.
raw_observation	log_id	TEXT	PK (composite); FK; Links to log.log_id.
raw_observation	project_id	TEXT	FK; Links to project.project_id.
raw_observation	source_id	TEXT	FK; Links to source.source_id.
raw_observation	period_type	INTEGER	not null; 1=single_year; 2=multi_year.
raw_observation	start_year	INTEGER	not null; Start year.
raw_observation	end_year	INTEGER	not null; End year.
raw_observation	raw_value	TEXT	not null; Source or documented compiler amount, stored as text.
raw_observation	raw_unit	TEXT	not null; Controlled mass unit: Mt, kt or tonnes.
raw_observation	priority_tier	INTEGER	not null; 1=selected for headline processing; 2=retained alternative. Independent of source provenance.
raw_observation	standards_text	TEXT	nullable; Standards cited (if any).

Table	Field	Type (SQLite)	Notes / constraints
raw_observation	raw_observation_notes	TEXT	nullable; Page/table refs; assumptions.
raw_observation	is_double_checked	INTEGER	not null; 0/1; four-eyes review flag (added v5).
raw_observation	is_storage_data_fixed	INTEGER	not null; 0/1; value confirmed final after review (added v5).
raw_observation	metric_basis	TEXT	nullable; program_net / net_stored / gross_injected / co2_received / not_stated; added in v8; coverage as specified in Section 4.7.4
raw_observation	metric_basis_notes	TEXT	nullable; source evidence and locators for the metric-basis classification, with page/sheet/cell references (added in data release v8)

C.4 Table exports and joined queries.

The notebook exports the four core tables directly from SQLite, preserving every field. Analysts can join raw_observation to log, project and source using the released keys or query v_observation_full for a flat view. The provided query script targets SQL Server; the notebook reads the SQLite views. Project-year transformations are performed by v_headline_series, not by the table-export step.